* What is Forensic Image of a device ?

A forensic image file is a bit-by-bit (sector-by-sector) copy of a storage device saved as a file.

Eg.
Original HDD / USB → Evidence.E01

It can contain :

- Existing files
- Deleted files
- File-System information
- Unallocated Space
- Partitions
- Sometimes hidden data.

- The important point is that it copies more than just the visible files.
- It attempts to preserve the storage device's data exactly as it existed.
- It preserves data integrity. [MD5, SHA1, SHA256]

* Why use Forensic Image ?

Investigators usually don't analyze the original device directly. They create an image and analyze the image instead, protecting the original evidence from modification.

* What is the difference betn cloning and imaging ?

Imaging	Cloning
1) Creates a forensic image file of a storage device.	1) Creates an exact copy directly onto another storage device.
2) Usually stored as .E01, .dd etc.	2) Usually copy sector by sector in another disk.
3) Can store matadata, compression & hashes.	3) Primarily produces a duplicate physical disk.

* MD5 gives o/p of 32 digit unique Code.
* SHA1 gives o/p of 40 digit unique Code.

* The image of device is smaller or equal to the size of that device.
  Eg. 20 GB pen drive
  Image ≤ 20 GB
- Sometimes the Compression reduce the image size

* Write Blocker –
  It Blocks the write function on the device & only allow to read the device.
  → H/w & S/w.

→ H/w & S/w

→ It blocks the write permission of that perticular device.

→ H/w ⇒ 1) Tableau Write Blocker TX1
It blocks the write permissions and also creates the image of that device.

2) Tableau Write Blocker TD4
It is a portable write Blocker.

FTK IMAGER

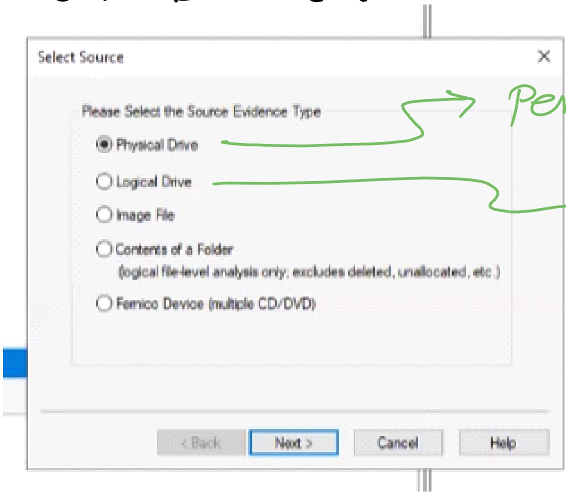
26 August 2026 12:17 PM

FTK => Forensic Tool kit

* How to use FTK Imager ?

St 1 - Files
↳ Create Disk Image.

St 2 - Select Desired



→ Pen drive, Hard disk etc.

→ Partition

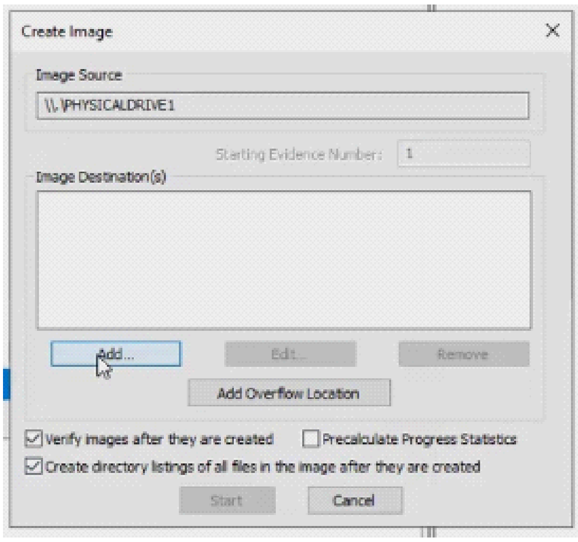
St 3 - If you have selected

1) physical → Choose the HDD/PD accordingly.
2) logical ⇒ Choose the partition.
3) Image ⇒ Browse the Image.
4) Content of folder ⇒ Browse the folder.

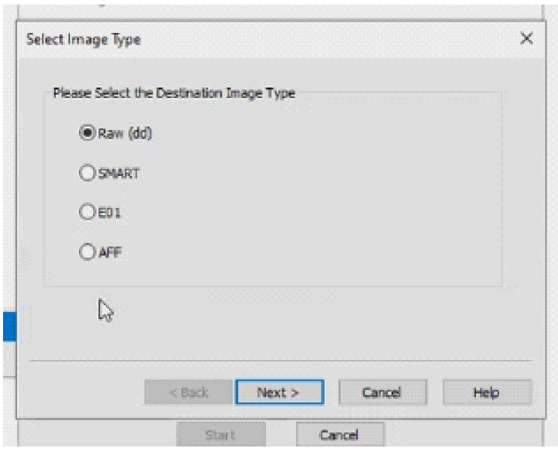
etc.

Remember - In physical we can get data recovery option but in logical not.

St 4 - Click Add



St 5 – Select format of Image.



i) Row (dd) – In Image No checksum, No Compression, No metadata content. But it is fast

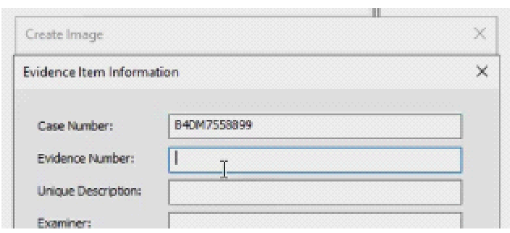
ii) SMART – Used for Linux.

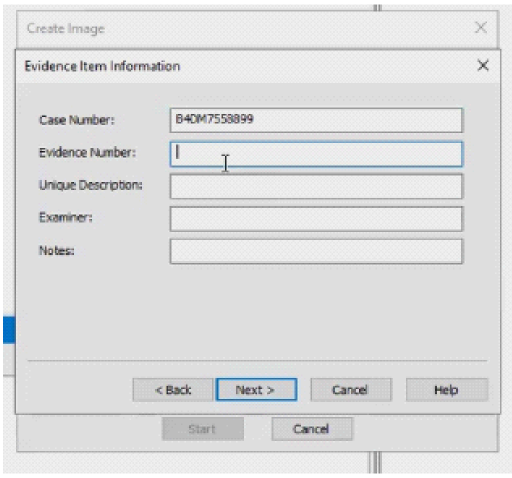
iii) E01 – In image we get Checksum, compression, matadata. So its slow.

iv) AFF – (Advance Forensic File)

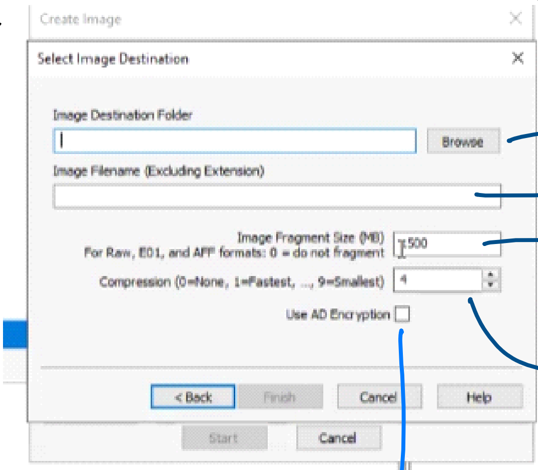
→ Default Market standards are E01 & dd. You can use it in Autopsy etc tools.

St 6 – Fill this Information





St7 – Destination of Image (Browse)



location to store Image

Image Name

Each Fragment / Packet Size

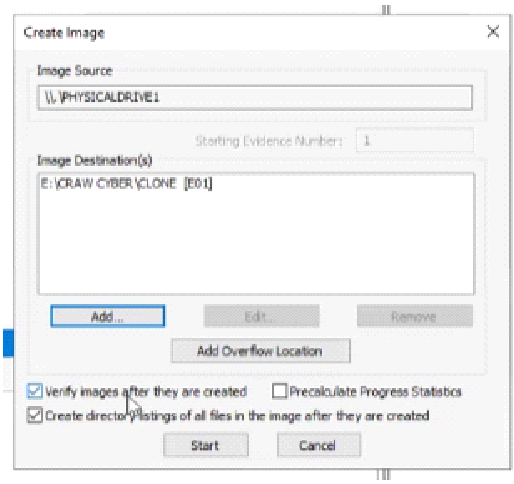
Compress the Size

0 => No Compression

9 => Smallest Compression

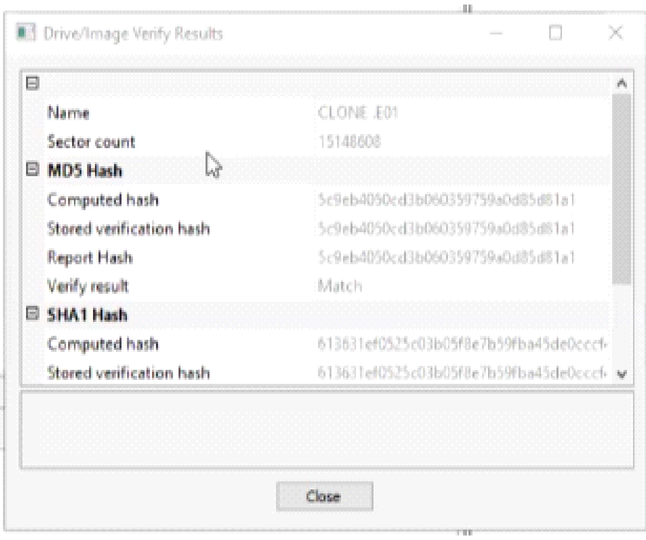
Encryption

St8 – Start



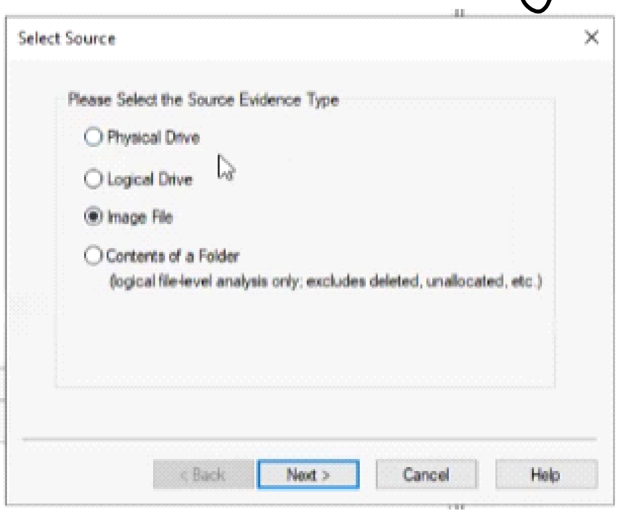
St9 – After Completion

St 9 - After Completion we get hash values.



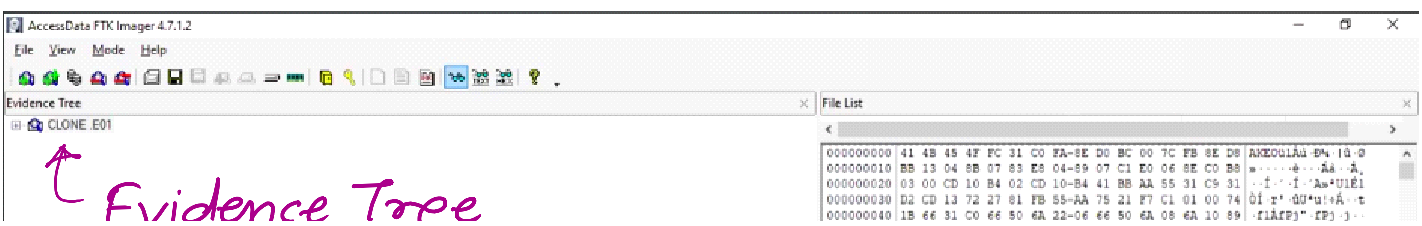
* To Check the image →

St 1 - Select the image file option

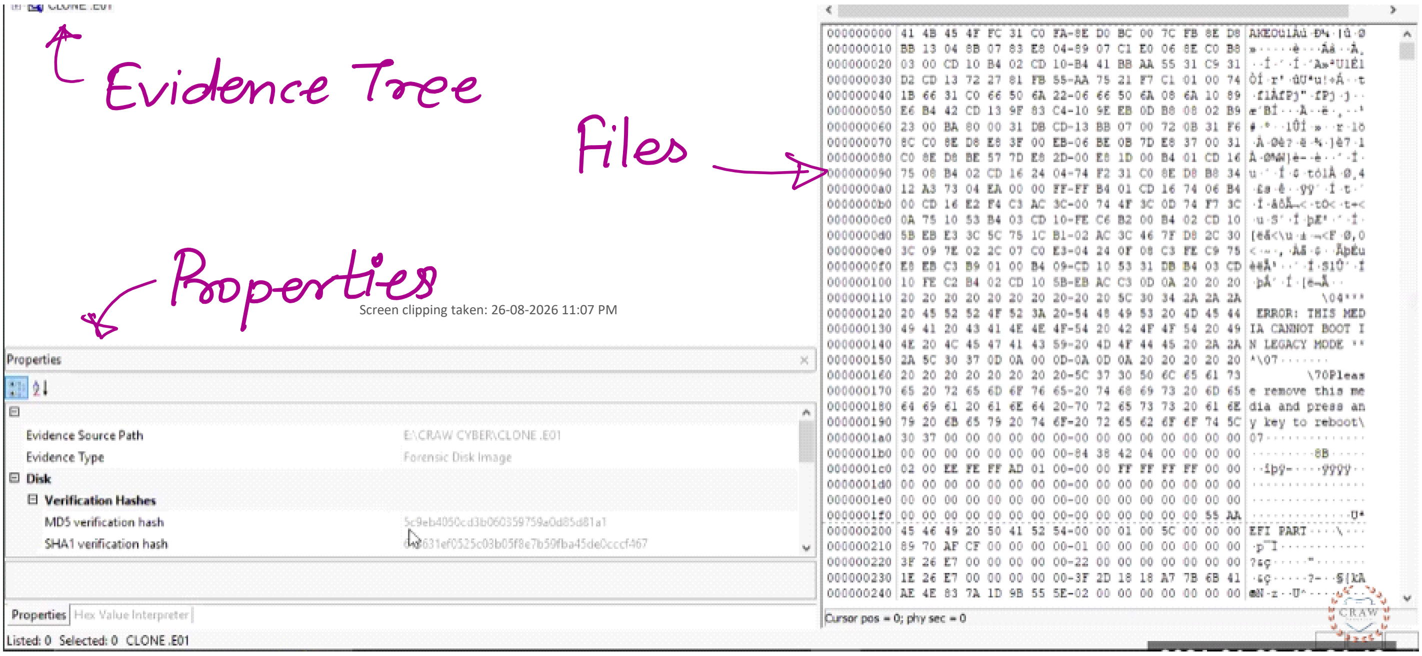


St 2 - Browse the image file.

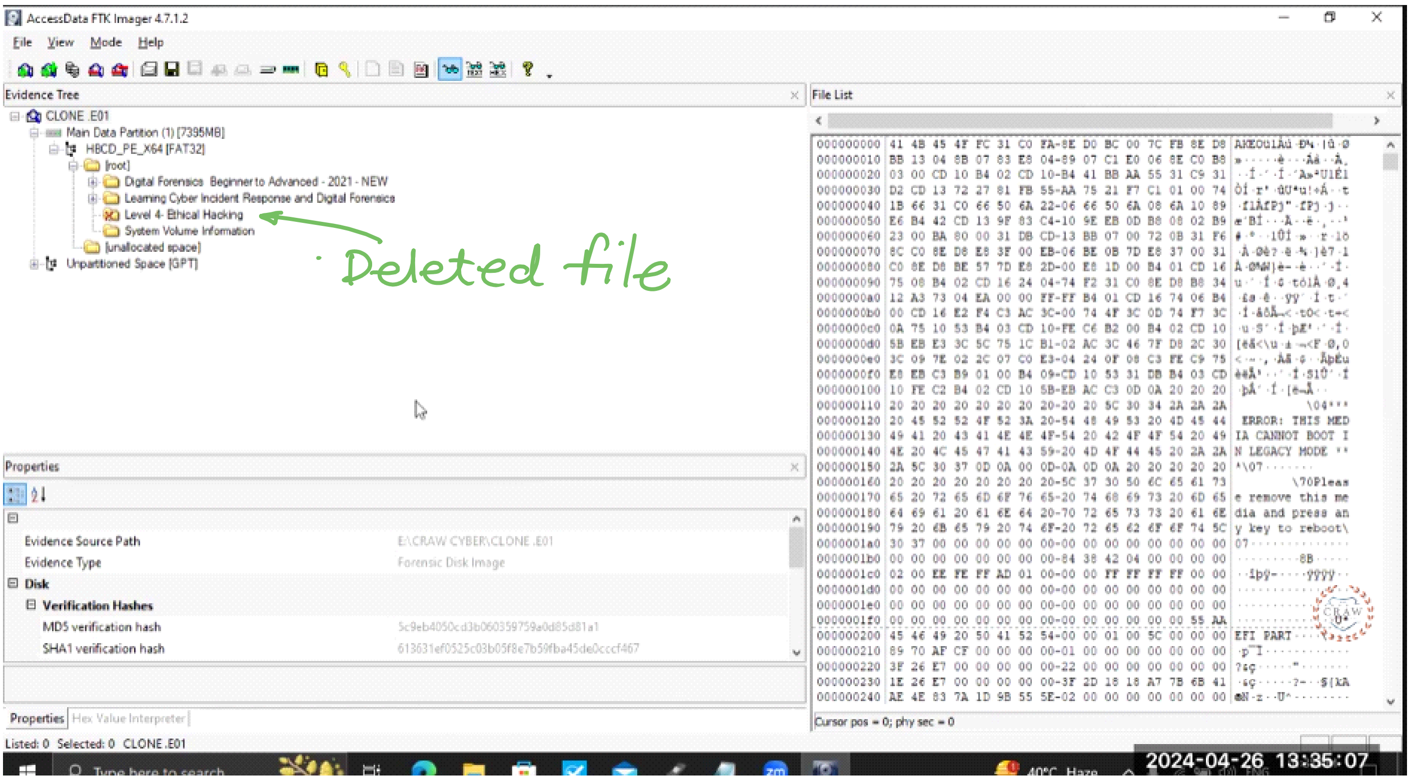
St 3 - See the information



Evidence Tree

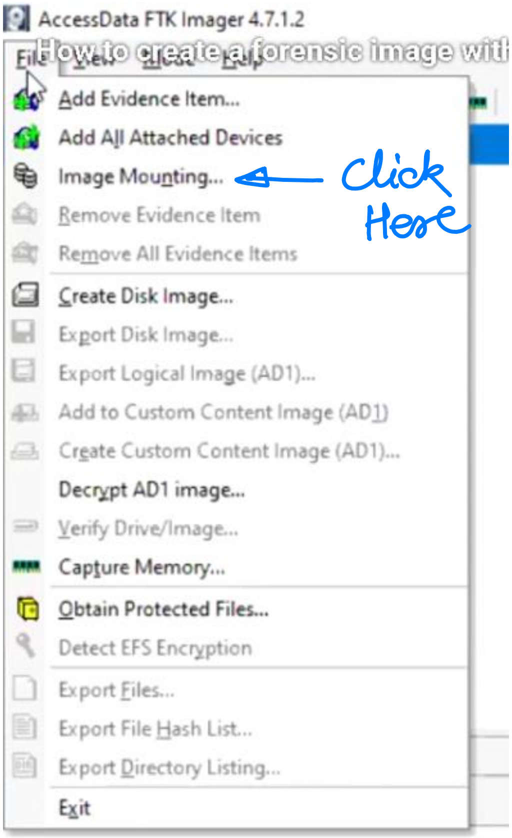


Deleted Data —

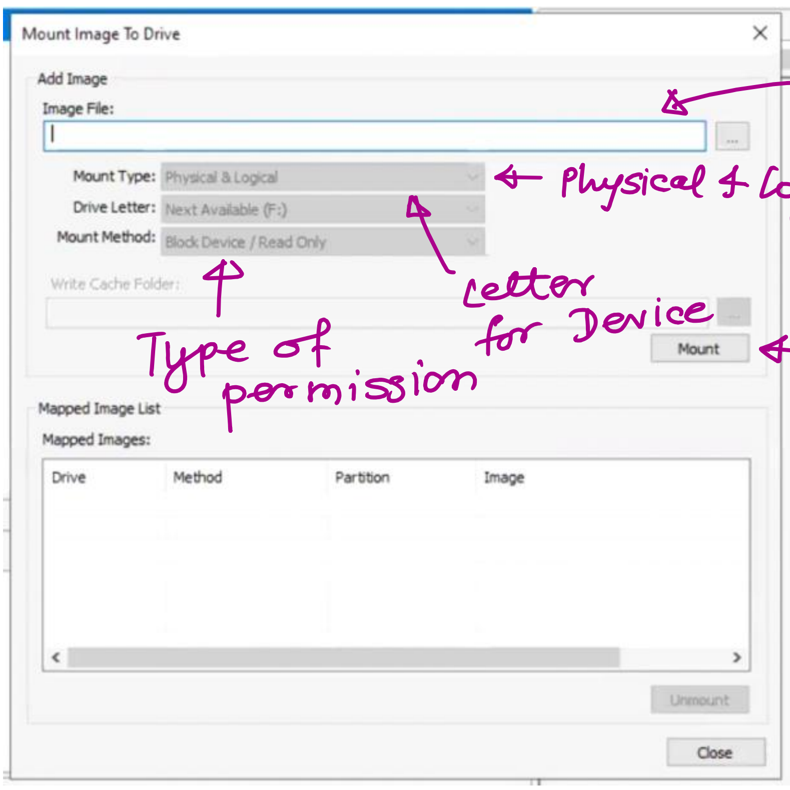


Mount Physically & logically —

St 1 -



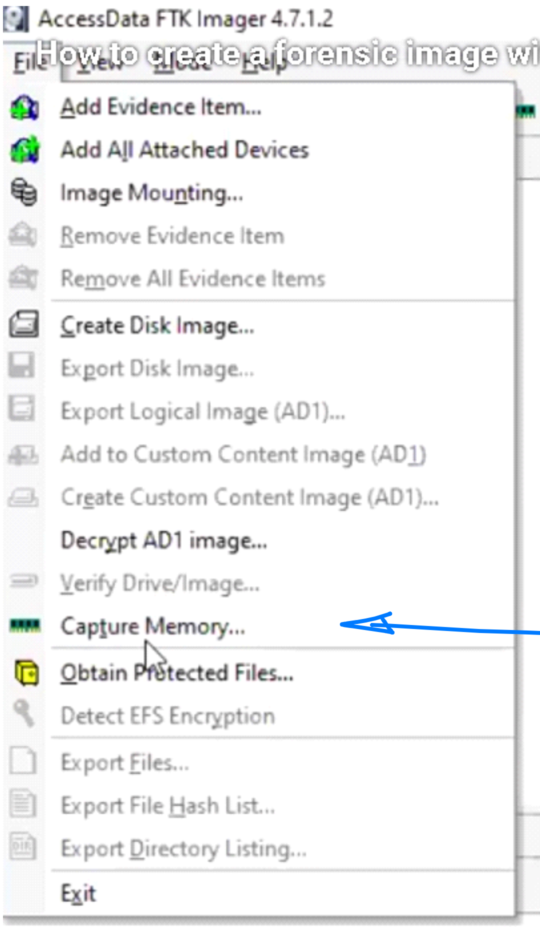
Step 2 –



Step 3 – You can see your device in the file manager.

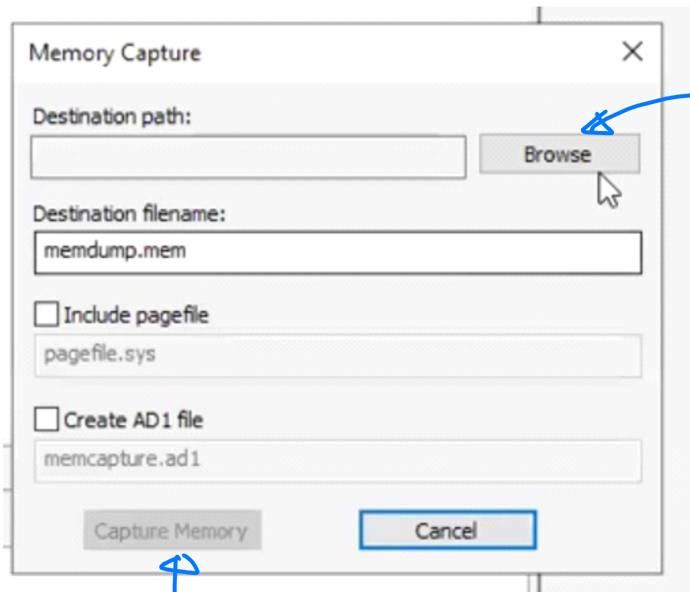
* Dump the RAM of the Evidence →

Step 1 –



click Here

Step 2 –



Give path where you want to store the memory

Click on Capture Memory

Click on Capture Memory.